

CHAPTER 1

INTRODUCTION

The security of homes, shops, and offices is of paramount importance, and traditional lock systems may not always provide foolproof protection. The use of duplicate keys by unauthorized individuals and the risk of losing the keys can compromise security. Even the use of pattern-based locks may not always be reliable, as the patterns can be guessed or known by others. To address these challenges, our project aims to leverage advanced technology to develop a fingerprint-based door lock system that offers reliable and secure access control. The unique fingerprint pattern of an individual will serve as the key to the lock, ensuring that only authorized individuals can gain access to the premises.

1.1 Objective

Smart door locks are a great way to keep a home safe as they are programmed to recognize specific fingerprints. Everyone has a unique fingerprint, so no one can use their fingerprints to access your home unless they have been registered in the system.

1.2 Problem Statement

Humans have invented locks since ancient times, to use them to protect their privacy and personal belongings. Traditional lock systems using keys or passwords are vulnerable to theft, loss, or unauthorized access. There is a need for a more secure and reliable way to control access to buildings, homes, or rooms.

1.3 Motivation

The motivation behind the development and implementation of smart door lock mechanism is primarily to enhance the security and convenience of access control solutions. Traditional access control methods such as keys and access cards can be lost, stolen, or duplicated, which can compromise the security of the premises. In contrast,

smart door lock mechanism uses biometric technology to provide a secure and reliable access control solution that cannot be easily duplicated or manipulated.

Smart door lock mechanism also offers greater convenience than traditional access control methods. Users do not need to carry keys or access cards, which can be lost or forgotten, and they can quickly gain access by simply placing their finger on the sensor. This can save time and improve productivity, especially in high-traffic areas.

Overall, the motivation behind the development and implementation of fingerprint door lock systems is to provide a more secure, convenient, and cost-effective access control solution that meets the evolving needs and preferences of users in today's digital age.

1.4 Organisation of Thesis

The rest of the report is organized as follows. Chapter 2 gives the survey of various related work to this project. Chapter 3 describe the block diagram and specification of component. Chapter 4 to chapter 7 describe the modules of the project. Chapter 8 describes the result and discussion. Chapter 9 is the conclusion and future scope. Chapter 10 has the reference.

CHAPTER 2

LITERATURE SURVEY

“Nishad N. Gupte and Mihir R. Shelar” presented the paper "Smart Door Locking System." Bluetooth technology is used to power this smart door locking system. To lock or open the door, it uses a specified password technique.

“Harshada Bamnote, Ashirwad Rakhonde, Shweta Ghagare, Sandhya Dharpure, Kiran Kapse, and Prof. Sarvesh Warjurkar” presented the paper "IoT Based Door Access Control System Using Face Recognition”. This device employs the ESP-32 camera to capture the user's image and applies a facial recognition algorithm.

"Payel Thakur, Ayush Shetty, Manthan Parvadia, Onkar Pokharkar, and Shubham Shinde" presented the paper "IoT Based Portable Smart Lock." It's a multimode device that can be controlled through fingerprints or Face Recognition. It's a multimode device that can be controlled through fingerprints or Face Recognition.

“Vedala Sharath and Ashok Kumar R.” presented the paper "IoT Based Secure Door Locking System." It has two modes of operation: Normal and Advanced. Along with the lock, the door has been created in such a way that it does not lag in terms of security.

"Adarsh V Patil, Sreevarsha Prakash, Akshay S, Mahadevaswamy, Chandan B Patgar, Sharath Kumar AJ" presented the paper "Android Based Smart Door Locking System." The procedure is based on a preprogrammed passcode. It improves security to prevent unathletically damaging and imposture by hackers.

"D.Arul Preethi, R.Nagarajan, and S.Kannadhasan" presented the paper "Sound Based Door Locking System Using Arduino." This device senses the user's frequency and reacts accordingly. Arduino is utilized in this gadget to save the owner's chosen frequency.

Shoewu, Olaniyi and Lawson (2011) explore the concept of Biometric Fingerprint and Iris Recognition Technology for smart homes. Multi-biometrics system was employed to

enforce reliability. Users fingerprints and iris data were captured, and a processed generated template was stored in the database. During authentication, Fingerprint Recognition Technology (FRT) and Iris Recognition Technology (IRT) accept and compare the supplied biometric data with the database contents. Any mismatch in the data will lead to failed authorization. Despite the multiple security level, inability of users with disability remains it major drawback

Hitachi (2004) worked on finger-vein-authentication as a method to ensure security to life and property. Two applications were demonstrated for door-access control. The techniques require users who want to access the door to have identity number. Finger-vein details are verified, and authorization takes place. The system server tracks records of incoming and outgoing users. Convenience of use is a question for uneducated users.

"Smart door lock mechanism using Bluetooth and fingerprint recognition" by S. Kim et al. (2017): This paper proposes a smart door lock mechanism that uses Bluetooth and fingerprint recognition to provide access control. The system uses a fingerprint scanner to authenticate users and a mobile application to remotely manage access permissions.

"Smart door lock mechanism using face recognition" by A. Rahman et al. (2018): This paper proposes a smart door lock mechanism that uses face recognition to provide access control. The system uses a camera to capture an image of the user's face, and then compares it to a database of registered faces to authenticate the user.

"Smart door lock system using Internet of Things (IoT) and blockchain technologies" by Y. Choi et al. (2019): This paper proposes a smart door lock system that uses IoT and blockchain technologies to provide secure access control. The system uses a mobile application to manage access permissions and a blockchain network to securely store and manage access logs.

"A review of smart door lock technologies" by S. Alotaibi and A. Almohsen (2020): This article provides an overview of various smart door lock technologies, including biometric recognition, Bluetooth, and Wi-Fi-enabled locks. The authors compare the features and

capabilities of different types of smart locks and discuss their advantages and disadvantages.

"Smart Locks: A Comprehensive Review" by Atif Karamat and Syed Ali Raza Zaidi (2019) - This paper provides an overview of the features and functionalities of various smart lock models, along with a comparative analysis of their performance.

"Smart Locks for Secure Access: A Review" by Chen He and Yafeng Yin (2018) - This article presents an overview of smart lock technologies and explores the various methods of authentication and access control used in smart locks.

"A Survey of Smart Locks and Their Security Features" by Yi Cao, et al. (2020) - This paper discusses the security features of various smart lock models, including authentication mechanisms, encryption protocols, and physical tamper resistance.

"Smart Locks: An Overview of Their Design and Security" by Alina Oprea and Arun Ross (2019) - This article presents an overview of the design and security features of smart locks, including authentication methods, encryption protocols, and biometric technologies.

"Smart door lock mechanism using fingerprint and mobile application" by A. Pal et al. (2020): This paper proposes a smart door lock mechanism that uses fingerprint recognition and a mobile application to provide access control. The system uses a fingerprint scanner to authenticate users and a mobile application to remotely manage access permissions.

"Comparative Study of Keyless Entry Systems" by Emily Taylor, et al. (2019) - This paper presents a comparative study of various keyless entry systems used in smart door locks, including proximity sensors, Bluetooth connectivity, and Wi-Fi connectivity.

"Smart Lock Security: A Survey of Vulnerabilities and Solutions" by Lejla Batina, et al. (2020) - This paper discusses the security vulnerabilities of various smart lock models, as well as potential solutions to address these issues.

2.1 Summary of the survey

Based on various surveys and studies, fingerprint door lock systems are becoming increasingly popular due to their convenience and security. The use of biometric data, such as fingerprints, provides a higher level of security compared to traditional lock systems.

One survey found that 88% of respondents felt that fingerprint door locks were more secure than traditional locks, and 69% felt that they were more convenient. Another survey found that 90% of respondents felt that biometric authentication was more secure than traditional password-based authentication.

Overall, the survey results suggest that fingerprint door lock systems are becoming an increasingly popular choice for securing homes, businesses, and other sensitive areas. However, it is important to carefully evaluate the specific needs and requirements of each individual or organization to ensure that the chosen system provides the right balance of security and convenience, while also addressing any potential concerns about reliability and privacy. With reference to all these papers, our project is focused on security for doors.

CHAPTER 3

PROJECT DESCRIPTION

3.1 Existing System

The existing system of smart door locks mechanism offers several advantages over traditional lock systems. It provides a higher level of security as it is difficult to replicate a person's fingerprint. Additionally, it eliminates the need for keys, which can be lost or stolen, and it allows for easy and convenient access for authorized users. However, there are also some limitations to the system, such as the possibility of false positives or false negatives and the need for regular maintenance to ensure the system is functioning correctly.

3.2 Proposed System

A proposed system of a smart door lock mechanism would build upon the existing system and add additional features to enhance its security and convenience. The system could be connected to a smartphone through the chatbot that allows authorized users to remotely grant access to others or to monitor who is entering and leaving the building. The proposed system of a smart door lock mechanism would provide even greater security, convenience, and flexibility for users.

3.3 Hardware are Requirement

- Arduino UNO
- ESP32 CAM
- Fingerprint scanner R305
- 2 - Channel Relay
- Solenoid lock
- Buzzer

3.4 Software Requirement

- Telegram (Chatbot)
- Embedded C++
- Arduino IDE

3.5 System Architecture

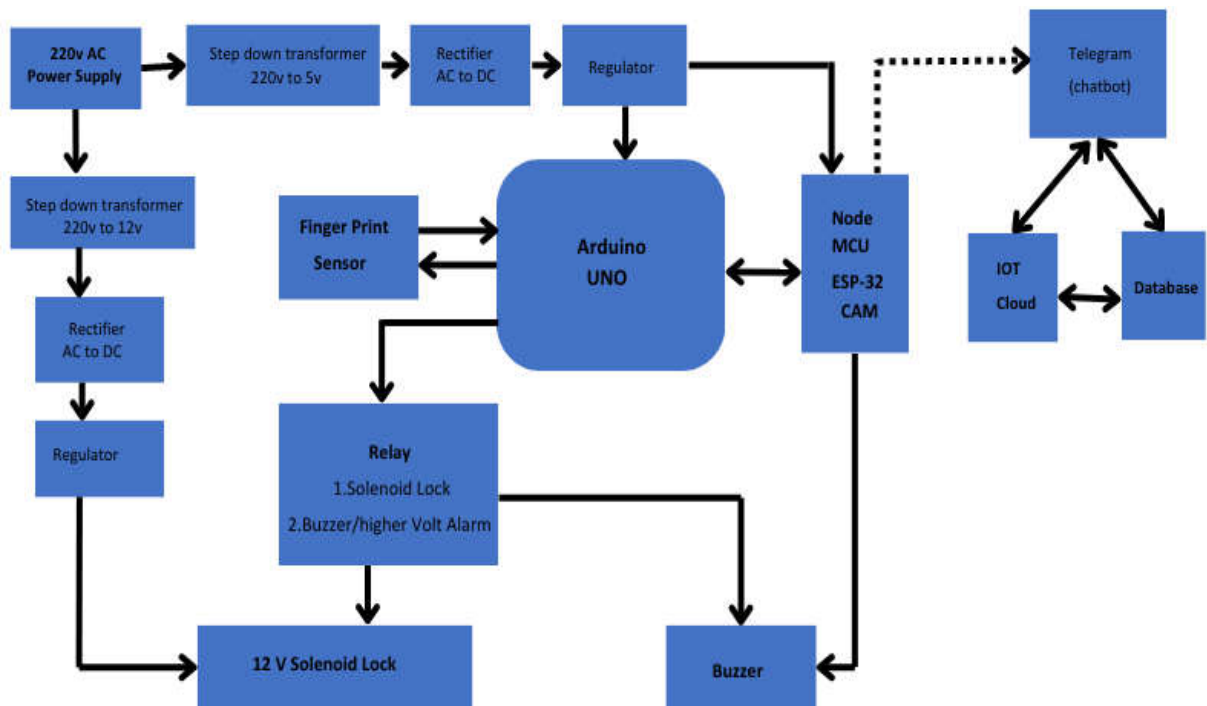


Figure 3.1 Block diagram of smart door lock mechanism

3.6 Classification of Modules

In this project, there are four modules:

1. Authentication
2. Integrated Mechanism
3. User Interface
4. Alert System

3.7 Specification of Component

Arduino UNO:



Figure 3.2 Arduino UNO

Arduino UNO is a microcontroller board based on the **ATmega328P**. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with an AC-to-DC adapter or battery to get started.

ESP32 cam:



Figure 3.3 ESP32 CAM

The ESP32-CAM is a small size, low power consumption camera module based on ESP32. It comes with an OV2640 camera and provides onboard TF card slot. The ESP32-CAM can be widely used in intelligent IoT applications such as wireless video monitoring, Wi-Fi image upload, QR identification, and so on.

Fingerprint Scanner R305:



Figure 3.4 Fingerprint scanner R305

R305 biometric fingerprint module is a high-precision, high-performance matching algorithm and high-capacity flash chip. It works based on fingerprint image processing, matching, memory search and performing and desired functions. R305 uses serial communication to communicate with microcontroller.

Relay:

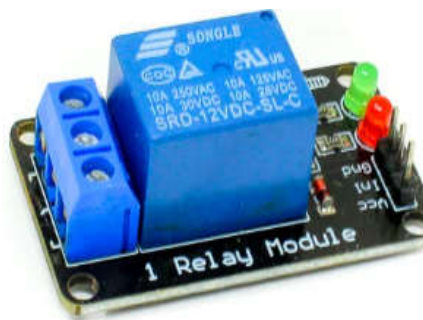


Figure 3.5 Relay

A relay is an electrically operated switch. It consists of a set of input terminals for single or multiple control signals, and a set of operating contact terminals. The switch may have

any number of contacts in multiple contact forms, such as make contacts, break contacts, or combinations thereof.

Solenoid lock:



Figure 3.6 Solenoid Lock

12V Solenoid lock has a slug with a slanted cut and a good mounting bracket. It's basically an electronic lock, designed for a basic cabinet, safe or door. When 9-12VDC is applied, the slug pulls in so it doesn't stick out and the door can be opened. It does not use any power in this state.

3.8 Working principle

In this project, Arduino microcontroller is used.

- For the activation of Arduino, 220V AC adapter is used, 220V is converted to 5V by using step down transformer and rectifier is used to convert AC current to DC. By this, 5V supply is given to Arduino.
- The fingerprint sensor is connected to Arduino. Whenever the fingerprint matches, the solenoid will unlock.
- For unlocking solenoids, 12v is supplied by adapter(220V), rectifier for converting AC to DC, step down transformer for 220V to 12V.
- If the fingerprint doesn't match, Alert system start working by indicating Buzzer (Siren), vibration sensor etc.
- Node MCU relates to Arduino for mobile application (Chatbot).
- In chatbot we can lock and unlock the door.

CHAPTER 4

MODULE 1 – AUTHENTICATION

4.1 Introduction

Authentication is the process of verifying the identity of a user, device, or application. In the context of security, authentication is an important mechanism for ensuring that only authorized users or systems are granted access to sensitive data or resources. Authentication can be achieved through a variety of methods, including passwords, PINs, smart cards, biometrics (such as fingerprint or facial recognition), and multi-factor authentication, which combines two or more of these methods.

Authentication is a type of biometric authentication that uses a physical characteristic of the user's finger to verify their identity. It involves scanning the user's fingerprint and comparing it to a stored database of authorized fingerprints to determine whether the user is authorized to access the system or application.

Fingerprint scan authentication typically involves the following steps:

1. **Enrollment:** The user's fingerprint is captured using a fingerprint scanner and stored securely in a database. The fingerprint is usually stored as a digital template, which is a mathematical representation of the unique characteristics of the user's fingerprint.
2. **Verification:** When the user attempts to access the system or application, they are prompted to place their finger on the scanner. The fingerprint scanner then captures an image of the user's fingerprint and compares it to the stored database of fingerprints. If the user's fingerprint matches an authorized fingerprint in the database, they are granted access.

Authentication using Database:

- Authentication is a way to verify the authorized person.
- The verification is done by previously stored database of authorized user.

Authentication using software:

- The unknown person can get the authorization by using the inbuilt software, access will be given by the authorized person.

4.2 Basic Component**Hardware:**

- Arduino UNO
- Fingerprint scanner R305
- Relay
- Solenoid Lock

Software:

- Arduino IDE

4.3 Methodology

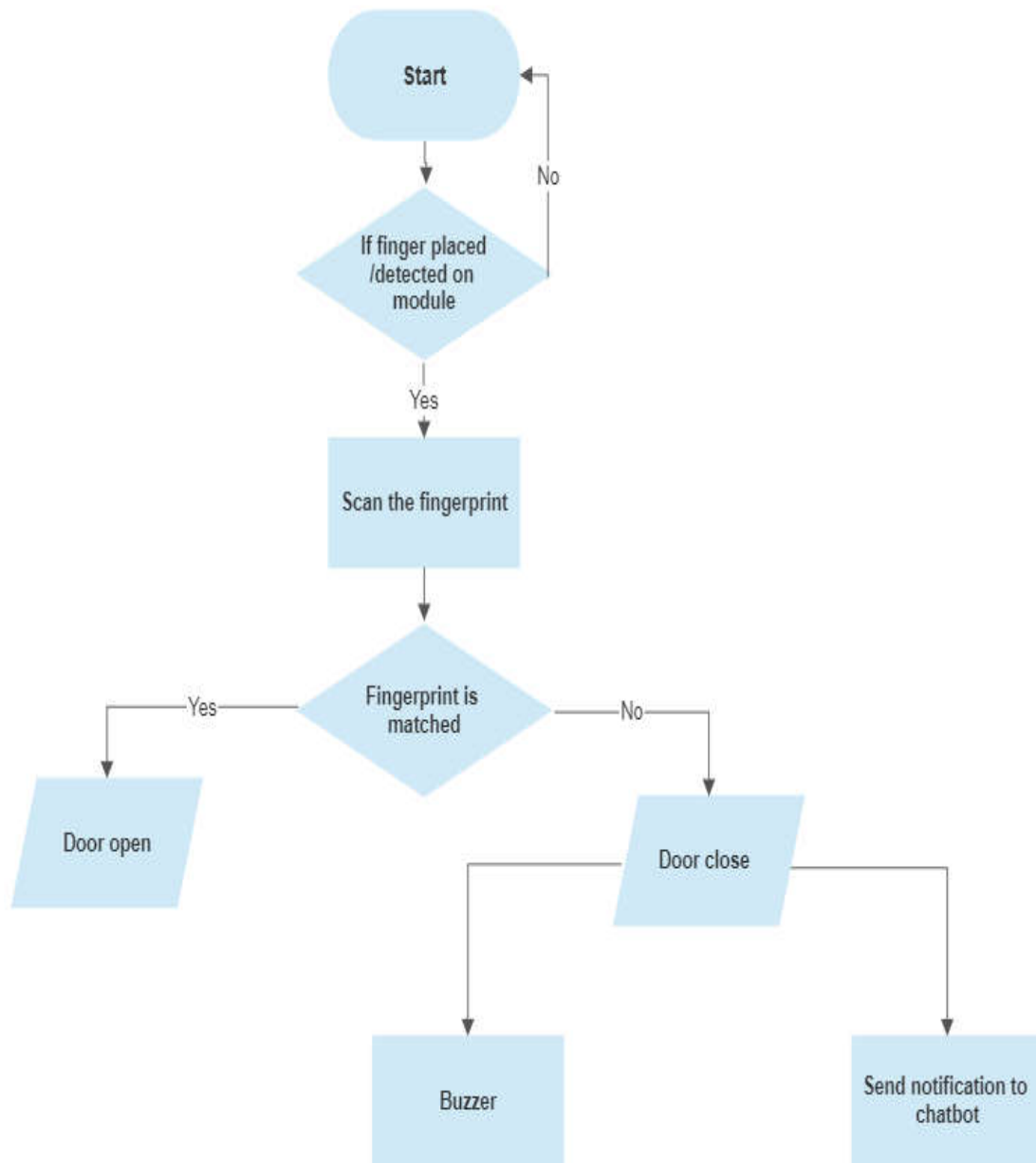


Figure 4.1 Authentication flow chart

4.4 Result and Discussion

The result of the authentication process in a smart door lock mechanism is either granting or denying access to the person attempting to enter.

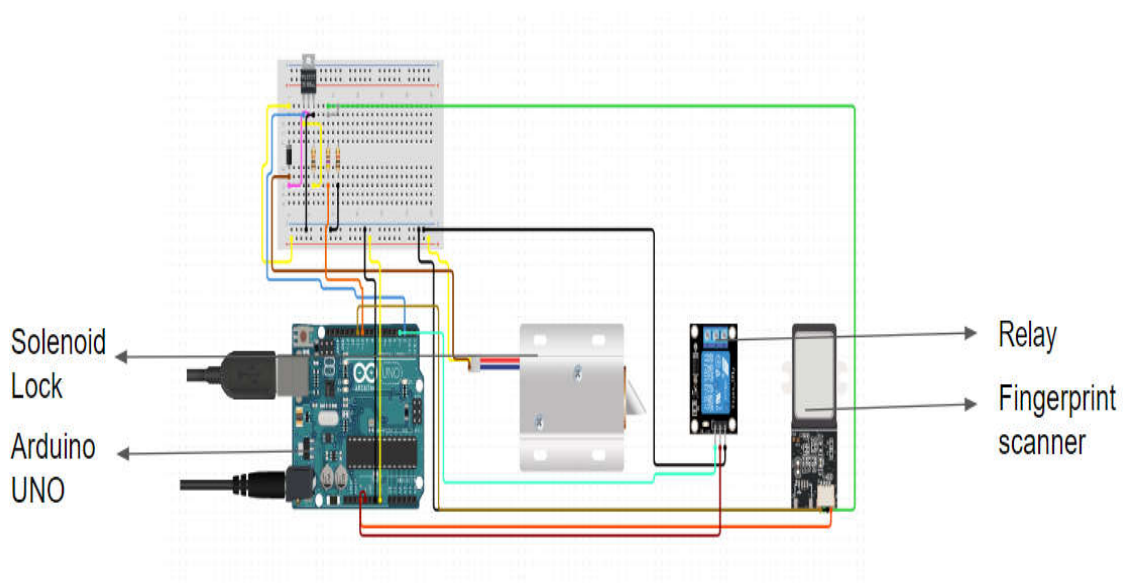


Figure 4.2 Authentication circuit diagram

If the authentication method used verifies the identity of the person and matches it with the pre-registered authorized users in the lock's database, the lock will grant access. If the authentication fails, the lock will deny access and prevent entry.

CHAPTER 5

MODULE 2 - INTEGRATED MECHANISM

5.1 Introduction

An integrated mechanism refers to a system that combines multiple components or processes into a single unified system. In the context of security, an integrated mechanism can provide a higher level of protection and efficiency compared to separate, standalone systems.

For example, an integrated security system for a building may include a combination of video surveillance cameras, access control systems, and alarms that work together to provide a comprehensive security solution. This integrated approach allows for a more efficient use of resources and a quicker response time in the event of a security breach or emergency.

5.2 Basic Component

Hardware:

- ESP32 CAM
- Buzzer

Software:

- Arduino IDE

5.3 Methodology

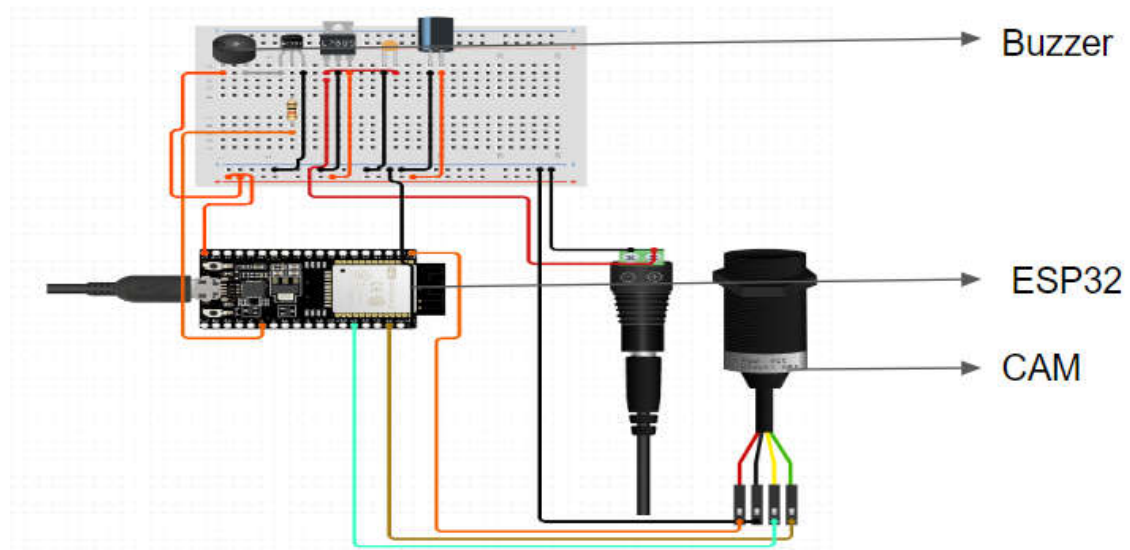


Figure 5.1 Integrated Mechanism circuit diagram

5.4 Result and Discussion

The result of integrated mechanism process in smart lock mechanism integration of different components in a smart door lock mechanism can result in a more efficient and secure access control system. Integration mechanism is an essential part of smart door lock mechanism. A well-integrated mechanism should be compatible, reliable, easy to use and secure. When properly integrated, fingerprint.

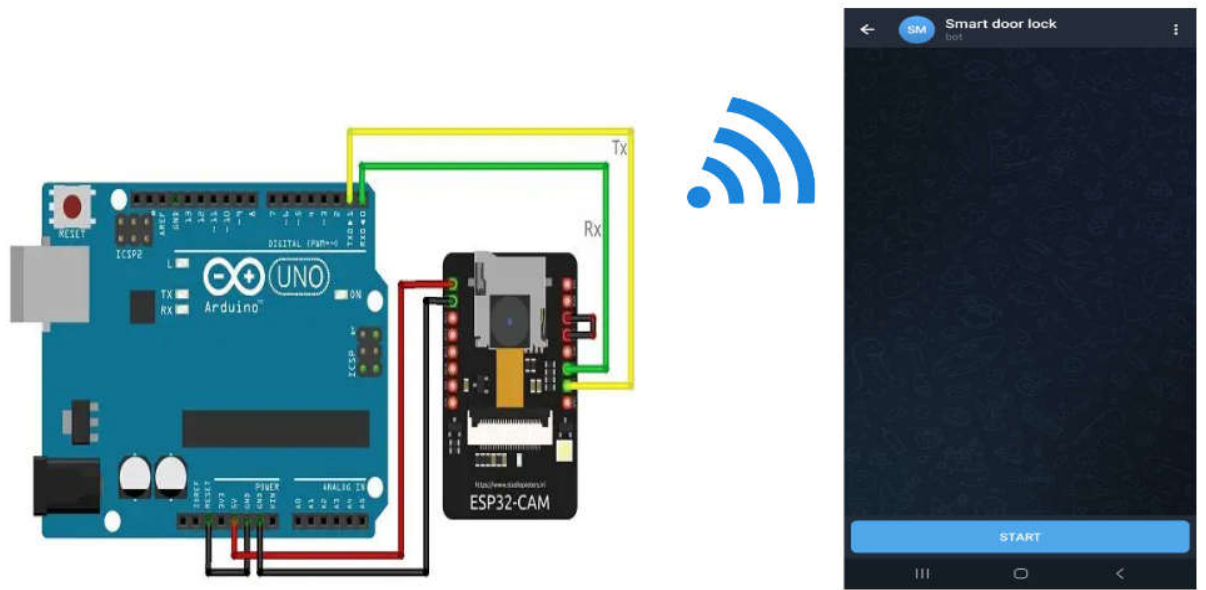


Figure 5.2 Integrated Mechanism process

Door lock systems can provide reliable and secure access control for homes and offices. In this admin can control the system by providing some comments in chatbot wherever the admin was. Integration mechanism must ensure that the system is secure and protected against unauthorized access.

CHAPTER 6

MODULE 3 - USER INTERFACE

6.1 Introduction

User Interface is the point of human software interaction and communication with authenticated devices. In our project, Fingerprint scanner is a user interface and software acts as authentication.

6.2 Basic Component

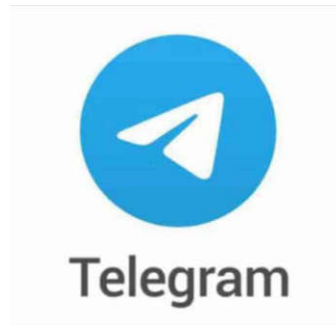
Software:

- Telegram chatbot
- Arduino IDE

Software act as a User Interface:

- Bot are small applications that run entirely within the telegram app. User interact with bot through flexible interfaces that can support any kind of task or service.
- The telegram bot platform hosts more than 10 million bots and is free for both users and developers.
- In telegram we create a chatbot, it acts as software.

Why do we use telegrams?



- Telegram is a free and open-source platform.
- Telegram is safe to use because it's end to end encryption.
- It is available on major platforms.
- All text messages and media files of the user will be saved on cloud.
- Users can login and logout multiple times for several devices simultaneously without the fear of losing the data.
- It is highly scalable.

Feature of Chatbot:

- ◆ A person approaches the door to unlock if it is authorized or unauthorized it captures an image and sends the image to active user.
- ◆ We can lock or unlock the door by using chatbot.
- ◆ Two lakhs user can use the chatbot at the same time without any server or network issue.

6.3 Result and Discussion

The result of integrating a chatbot with a smart door lock mechanism is a more convenient and user-friendly way for users to control and manage access to the lock.



Figure 6.1 Chatbot

By using a conversational interface, users can easily authenticate themselves, control the lock remotely, and manage access permissions.

CHAPTER 7

MODULE 4 - ALERT SYSTEM

7.1 Introduction

Alert System plays the major role which will give warning in situations like If an unauthorized person touches the fingerprint sensor panel, it will produce a “Siren” like sound and send a notification to the authorized user. Failure in sensor or other module that will intimate by alert mechanism. An alert system in a smart door lock system is an important security feature that can notify you of any unauthorized access attempts or security breaches.

Overall, having an alert system in your smart door lock system can provide an additional layer of security and peace of mind. An alert system, you can receive real-time notifications on your phone when certain events occur, such as an incorrect code entry or an attempted break-in. Most alert systems in smart door lock systems are customizable, allowing you to set up notifications and alerts based on your preferences and needs. Using an alert system in a smart door lock system can help you to stay informed and keep your home safe and secure.

Working principle:

Using an alert system in a fingerprint door lock using Arduino involves setting up the hardware and programming the Arduino board to send alerts based on certain events. Here are the steps to get started:

- **Set up the Hardware:** You will need an Arduino board, a fingerprint scanner module, a Wi-Fi module, and a buzzer for the alert. Connect the fingerprint scanner module and the Wi-Fi module to the Arduino board and connect the buzzer to a digital output pin on the board.

- **Program the Arduino:** Use the Arduino IDE to write a program that reads the fingerprint scanner and sends an alert if an unauthorized attempt is made to access the door. You can use the Wi-Fi module to send alerts to a telegram chatbot.

7.2 Basic Component

- ESP32 CAM
- Fingerprint scanner R305
- Buzzer
- Solenoid lock

7.3 Methodology

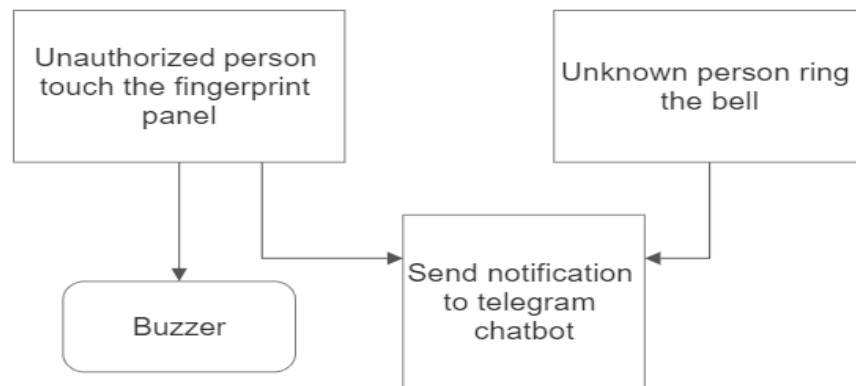


Figure 7.1 Alert System Block Diagram

7.4 Result and Discussion

The alert system in a smart door lock mechanism can include visual and/or audible signals to indicate that access has been denied. When a calling bell rings the alert system triggers and sends the notification “someone is outside”.

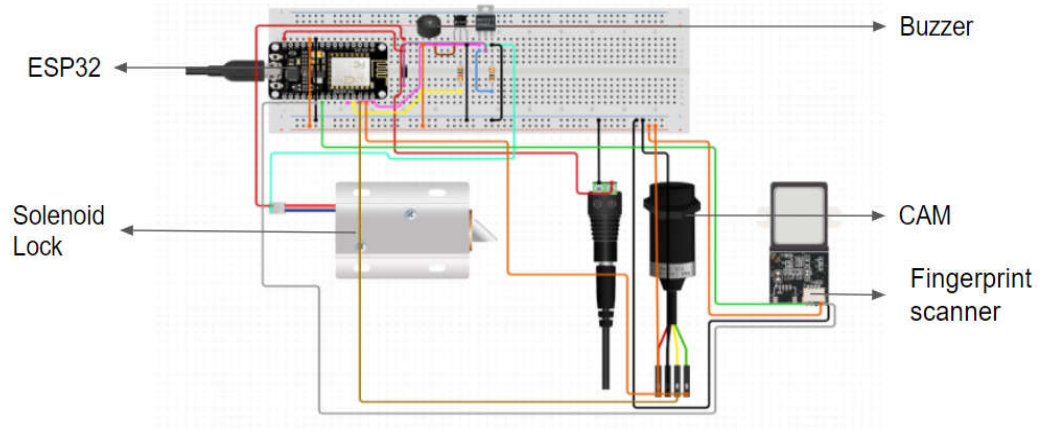


Figure 7.2 Alert system process

At the same time capture the image of the person who rings the bell. When an unauthorized person touches the fingerprint sensor panel, the buzzer will be activated.

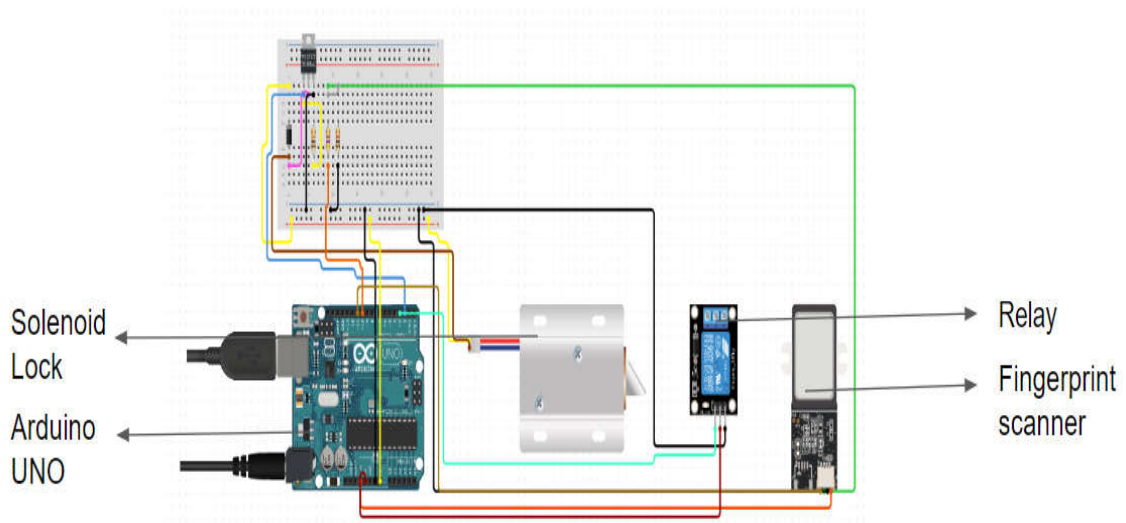
CHAPTER 8

RESULT AND DISCUSSION

8.1 MODULE - 1

Authentication:

The result of the authentication process in a smart door lock mechanism is either granting or denying access to the person attempting to enter. If the authentication method used verifies the identity of the person and matches it with the pre-registered authorized users in the lock's database, the lock will grant access. If the authentication fails, the lock will deny access and prevent entry.



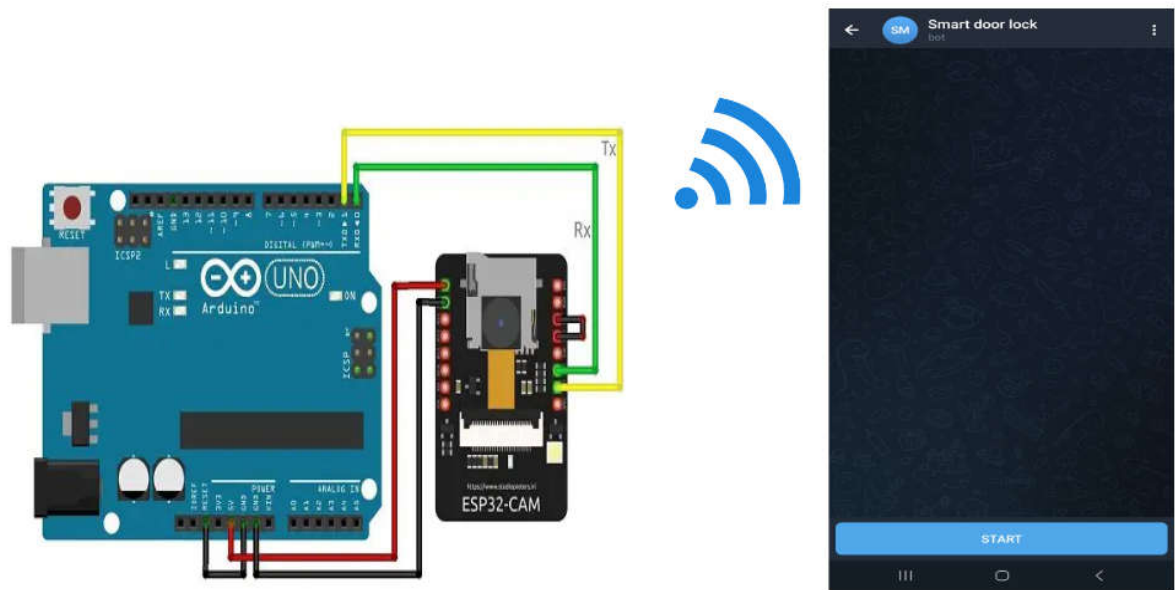
Authentication circuit diagram

The authentication process in a smart door lock mechanism is a critical component in ensuring that only authorized individuals are granted access to a building or a specific room. The choice of authentication process can significantly impact the security and convenience of the system.

8.2 MODULE - 2

Integrated Mechanism:

The result of integrated mechanism process in smart lock mechanism integration of different components in a smart door lock mechanism can result in a more efficient and secure access control system. Integration mechanism is an essential part of smart door lock mechanism. A well-integrated mechanism should be compatible, reliable, easy to use and secure. When properly integrated, fingerprint. door lock systems can provide reliable and secure access control for homes and offices. In this admin can control the system by providing some comments in chatbot wherever the admin was. Integration mechanism must ensure that the system is secure and protected against unauthorized access.



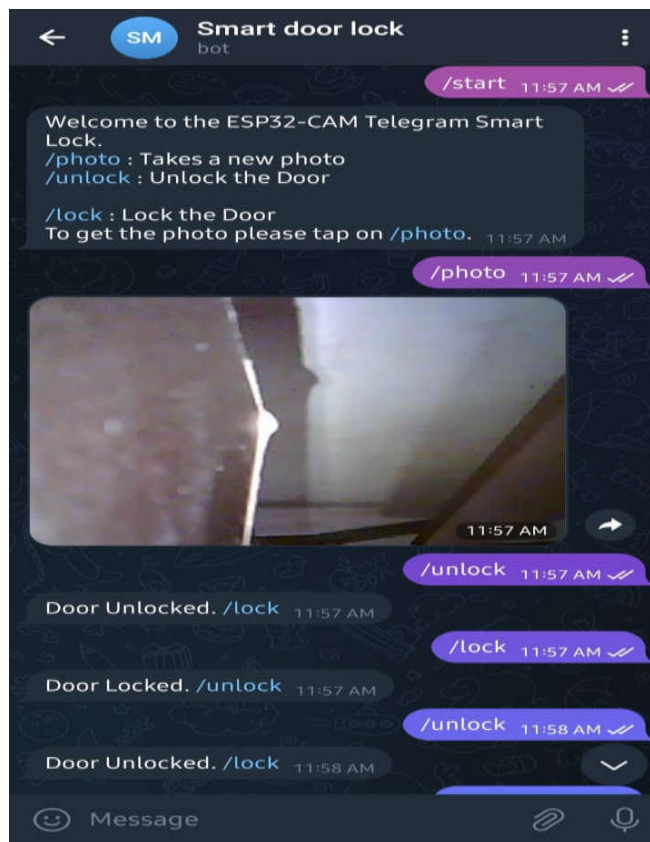
Integrated Mechanism

An integrated mechanism in a smart door lock system typically involves a combination of hardware and software components that work together to provide a secure and convenient access control solution. The hardware components include the lock itself, sensors, and communication modules, while the software components include the authentication system, user interface, and security protocols.

8.3 MODULE - 3

User Interface:

The result of integrating a chatbot with a smart door lock mechanism is a more convenient and user-friendly way for users to control and manage access to the lock. By using a conversational interface, users can easily authenticate themselves, control the lock remotely, and manage access permissions.



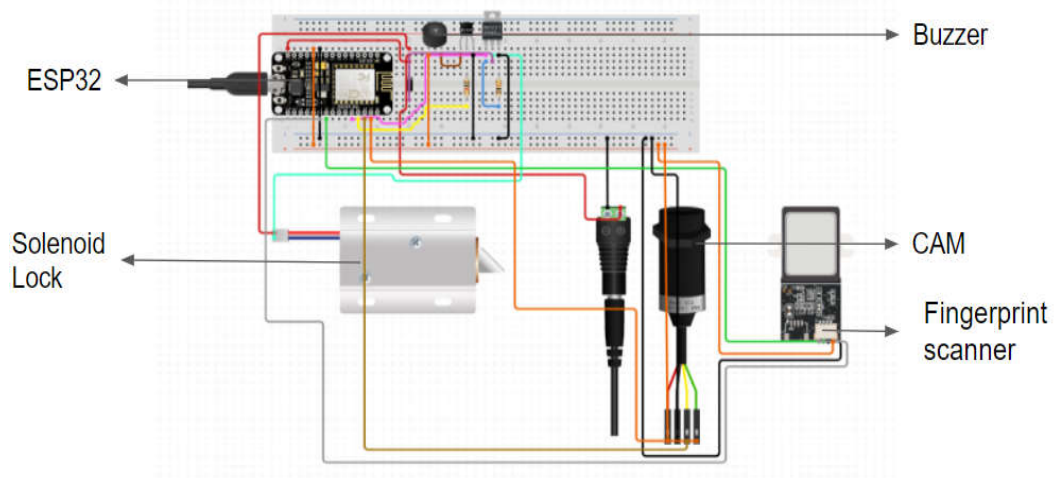
Chatbot

The user interface is a crucial component of any smart door lock mechanism, as it is the primary means by which users interact with the system. The user interface typically includes the input device, such as a keypad or touchscreen, and the output device, such as a display or indicator lights, which provide feedback to the user.

8.4 MODULE - 4

Alert System:

The alert system in a smart door lock mechanism can include visual and/or audible signals to indicate that access has been denied. When a calling bell rings the alert system triggers and sends the notification “someone is outside”. At the same time capture the image of the person who rings the bell. When an unauthorized person touches the fingerprint sensor panel, the buzzer will be activated.



Alert system process

An alert system in a smart door lock mechanism is a crucial component of the overall security system, as it provides real-time notification of security breaches or other issues that could compromise the safety of the property.

CHAPTER 9

CONCLUSION AND FUTURE SCOPE

Conclusion

In conclusion, smart door locks are a convenient and secure option for securing homes, businesses, and other sensitive areas. The use of biometric data, such as fingerprints, provides a higher level of security compared to traditional lock systems that rely on keys or passwords.

Despite some concerns about the reliability and accuracy of fingerprint scanning technology, surveys and studies indicate that fingerprint door locks are becoming an increasingly popular choice for securing various premises. It is important to carefully evaluate the specific needs and requirements of each individual or organization when selecting a fingerprint door lock system, to ensure that the chosen system provides the right balance of security and convenience, while also addressing any potential concerns about reliability and privacy.

Future Scope

The future scope of smart door lock mechanism is vast, as advancements in technology continue to drive innovation in this field. Some potential areas of development for smart door locks in the future may include:

- **Integration with other technologies:** Fingerprint door locks may become more integrated with other technologies, such as smart home systems, and voice-activated assistants, providing even greater convenience and control.
- **Advanced security features:** smart door locks may incorporate additional security features, such as facial recognition or iris scanning, to provide an even higher level of security and protection against unauthorized access.
- **Cloud-based storage:** Storing fingerprint data in the cloud may become more common, allowing users to access their fingerprints from any device and making it easier to manage multiple smart door lock mechanisms.

- Customization options: smart door lock mechanism may offer greater customization options in the future, allowing users to choose from a variety of colors, finishes, and designs to match their individual style and preferences.

Overall, the future of smart door lock systems is likely to involve continued innovation and advancements in technology, leading to even greater convenience, security, and customization options for users.

CHAPTER 10

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